Diabetes Classification POC using Spring Boot and Azure OpenAI

Objective: Develop a Spring Boot REST API that classifies diabetes risk based on patient profile using Azure OpenAI GPT-5-mini (Chat Model).

Technology Stack: - Java 17 - Spring Boot 3.5.6 - Azure OpenAI (gpt-5-mini) - Maven

Endpoints: - POST `/predict` - Accepts patient data and returns diabetes risk with patient profile.

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1. Sample Request JSON

```json
{
  "age": 50,
  "weight": 180,
  "height": 5.7,
  "glucose": 140,
  "bloodPressure": 85,
  "insulin": 90
}
```

2. Sample Response JSON

```json
{
  "patientProfile": {
    "age": 50,
    "weight": 180,
    "height": 5.7,
    "bmi": 29.39,
    "glucose": 140,
    "bloodPressure": 85,
    "insulin": 90
  },
  "diabetesRisk": {
    "riskLevel": "High",
    "probability": 0.78
  }
}
```

3. Flow Diagram

```
+--------------------+        +------------------------+
+--------------------+
|  Patient sends     | POST   |  Spring Boot REST API  |  -->   |
DiabetesService   |
|  JSON /predict     |------->|  /predict endpoint     |        |  processes
data,   |
+--------------------+        +------------------------+        |  calculates
BMI,   |
                                                                  |  calls Azure

OpenAI |
                                                                  |  Chat

model         |

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                                                                           |
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+--------------------+
                                                                  |  Azure

OpenAI      |
                                                                  |  GPT-5-mini

Chat   |
                                                                  |  Model

returns     |
                                                                  |  risk

JSON         |

+--------------------+
                                                                           |
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+--------------------+
                                                                  |  Service

combines  |
                                                                  |  patient

profile & |
                                                                  |  risk and

returns  |

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                                                                           |
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+--------------------+
                                                                  |  Response to
```

```
Client|

+--------------------+
```

4. Key Implementation Points

- Height input in feet, converted to meters for BMI calculation.
- Patient profile returned with BMI, glucose, blood pressure, insulin.
- Diabetes risk returned as JSON with `riskLevel` and `probability`.
- Uses **Azure OpenAI GPT-5-mini chat model** via `/chat/completions` endpoint.

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5. Sample Spring Boot Service Snippet

```java
// inside DiabetesService
if (patientData.getBmi() == 0 && patientData.getHeight() > 0) {
    double heightInMeters = patientData.getHeight() * 0.3048;
    double bmi = patientData.getWeight() / (heightInMeters * heightInMeters);
    patientData.setBmi(Math.round(bmi * 100.0) / 100.0);
}
```

```java
Map<String, Object> message = Map.of("role", "user", "content", prompt);
Map<String, Object> requestBody = Map.of("messages", List.of(message),
"max_tokens", 256);
RestTemplate restTemplate = new RestTemplate();
String rawResponse = restTemplate.postForObject(url, entity, String.class);
```

This POC demonstrates: - How to integrate Spring Boot with Azure OpenAI chat models. - How to process patient data and calculate BMI. - How to return structured risk prediction with patient profile.